

AC 间二层组网隧道转发无线漫游

一、实验目的

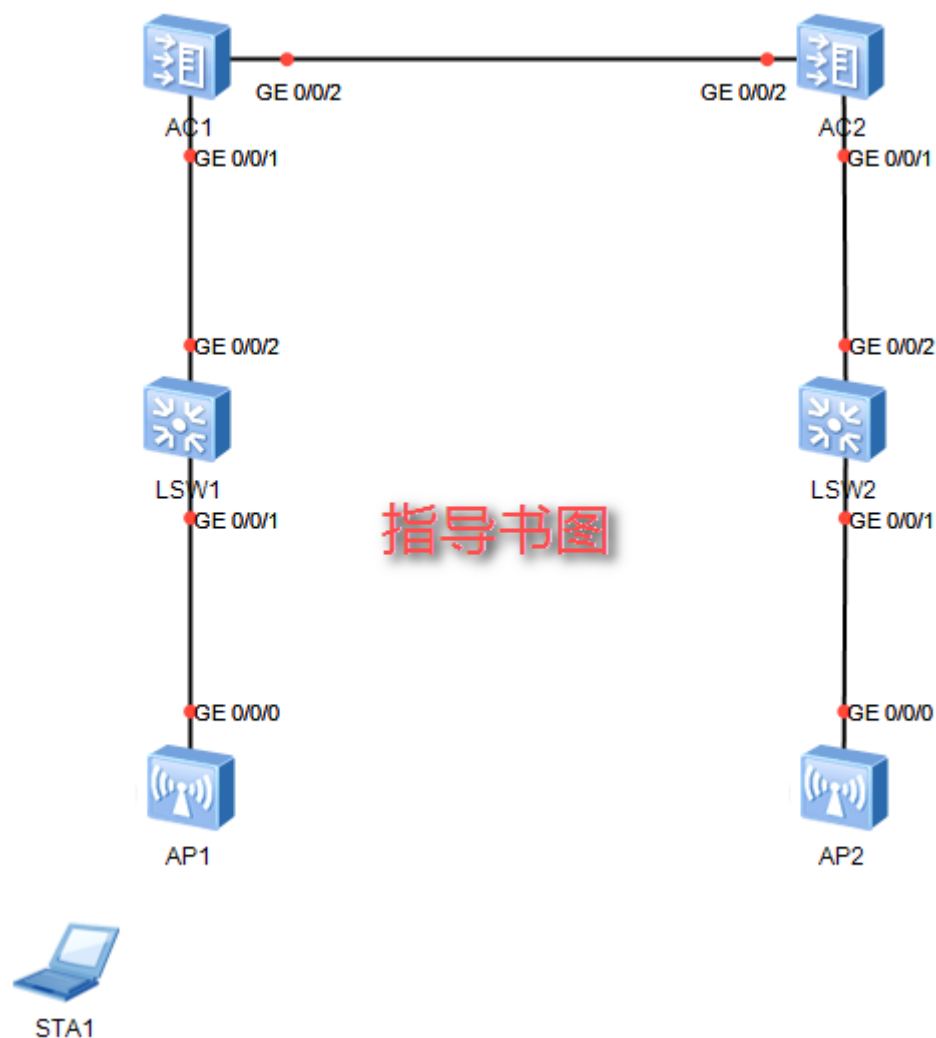
了解 AC 间二层组网隧道转发无线漫游相关知识，隧道转发的含义，配置方法。

二、模拟器版本

ensp1.3.00.100 版本

三、实验拓扑

3.1 网络拓扑



3.2 参数规划表

配置项	规划数据
管理 VLAN	VLAN100
STA 业务 VLAN	VLAN101
DHCP 服务器	AC_1 作为 DHCP 服务器为 AP 和 STA 分配 IP 地址
AP 的 IP 地址池	10.23.100.3 ~ 10.23.100.254/24 STA 的 IP 地址池
AC 的源接口 IP 地址	VLANIF100 AC_1: 10.23.100.1/24 AC_2: 10.23.100.2/24
AP 组	名称: ap1

	引用模板: VAP 模板 wlan-net、域管理模板 default
域管理模板	名称: default 国家码: 中国
	调优信道集合: 配置 2.4G 和 5G 调优带宽和调优信道
SSID 模板	名称: wlan-net
	SSID 名称: wlan-net
安全模板	名称: wlan-net
	安全策略: WPA-WPA2+PSK+AES
	密码: a1234567
VAP 模板	名称: wlan-net 转发模式: 隧道转发 业务 VLAN: VLAN101
	引用模板: SSID 模板 wlan-net、安全模板 wlan-net
空口扫描模板	名称: wlan-airscan 探测信道集合: 调优信道 空口扫描间隔时间: 60000 毫秒 空口扫描持续时间: 60 毫秒
RRM 模板	名称: wlan-rrm 信道自动调优: 开启 功率自动调优: 开启

四、实验步骤

4.1 根据网络拓扑在模拟器中搭建网络;

4.1.1 # 配置接入交换机 Switch_1 的 GE0/0/1 和 GE0/0/2 接口加入 VLAN100, GE0/0/1 的缺省 VLAN

为VLAN100。

```
<Huawei> system-view
[Huawei] sysname Switch_1
[Switch_1] vlan batch 100
[Switch_1] interface gigabitethernet 0/0/1
[Switch_1-GigabitEthernet0/0/1] port link-type trunk
[Switch_1-GigabitEthernet0/0/1] port trunk pvid vlan 100
[Switch_1-GigabitEthernet0/0/1] port trunk allow-pass vlan 100
[Switch_1-GigabitEthernet0/0/1] quit
[Switch_1] interface gigabitethernet 0/0/2
[Switch_1-GigabitEthernet0/0/2] port link-type trunk
[Switch_1-GigabitEthernet0/0/2] port trunk allow-pass vlan 100
[Switch_1-GigabitEthernet0/0/2] quit
```

4.1.2 # 配置接入交换机 Switch_2 的 GE0/0/1 和 GE0/0/2 接口加入 VLAN100, GE0/0/1 的缺省 VLAN

为VLAN100。

```
<Huawei> system-view
[Huawei] sysname Switch_2
[Switch_2] vlan batch 100
[Switch_2] interface gigabitethernet 0/0/1
[Switch_2-GigabitEthernet0/0/1] port link-type trunk
[Switch_2-GigabitEthernet0/0/1] port trunk pvid vlan 100
[Switch_2-GigabitEthernet0/0/1] port trunk allow-pass vlan 100
[Switch_2-GigabitEthernet0/0/1] quit
[Switch_2] interface gigabitethernet 0/0/2
[Switch_2-GigabitEthernet0/0/2] port link-type trunk
[Switch_2-GigabitEthernet0/0/2] port trunk allow-pass vlan 100
[Switch_2-GigabitEthernet0/0/2] quit
```

4.1.3 配置 AC

配置AC_1 的接口GE0/0/1 加入VLAN100, AC_1 的接口GE0/0/2 加入VLAN100 和VLAN101。

```
<AC6605> system-view
[AC6605] sysname AC_1
[AC_1] vlan batch 100 101
[AC_1] interface gigabitethernet 0/0/1
[AC_1-GigabitEthernet0/0/1] port link-type trunk
[AC_1-GigabitEthernet0/0/1] port trunk allow-pass vlan 100
[AC_1-GigabitEthernet0/0/1] quit
[AC_1] interface gigabitethernet 0/0/2
[AC_1-GigabitEthernet0/0/2] port link-type trunk
[AC_1-GigabitEthernet0/0/2] port trunk allow-pass vlan 100 101
[AC_1-GigabitEthernet0/0/2] quit
[AC_1] interface vlanif 100
[AC_1-Vlanif100] ip address 10.23.100.1 255.255.255.0
[AC_1-Vlanif100] quit
[AC_1] interface vlanif 101
[AC_1-Vlanif101] ip address 10.23.101.1 255.255.255.0
[AC_1-Vlanif101] quit
```

配置AC_2 的接口GE0/0/1 加入VLAN100, AC_2 的接口GE0/0/2 加入VLAN100 和VLAN101。

```
<AC6605>system-view
[AC6605] sysname AC_2
[AC_2] vlan batch 100 101
[AC_2] interface gigabitethernet 0/0/1
[AC_2-GigabitEthernet0/0/1] port link-type trunk
[AC_2-GigabitEthernet0/0/1] port trunk allow-pass vlan 100
[AC_2-GigabitEthernet0/0/1] quit
[AC_2] interface gigabitethernet 0/0/2
[AC_2-GigabitEthernet0/0/2] port link-type trunk
[AC_2-GigabitEthernet0/0/2] port trunk allow-pass vlan 100 101
[AC_2-GigabitEthernet0/0/2] quit
[AC_2] interface vlanif 100
[AC_2-Vlanif100] ip address 10.23.100.2 255.255.255.0
[AC_2-Vlanif100] quit
[AC_2] interface vlanif 101
[AC_2-Vlanif101] ip address 10.23.101.2 255.255.255.0
[AC_2-Vlanif101] quit
```

配置DHCP 服务器为STA 和AP 分配IP 地址

在AC_1 上配置VLANIF100 接口为AP 提供IP 地址, VLANIF101 接口为STA 提供IP 地址。

```
[AC_1] dhcp enable
[AC_1] interface vlanif 100
[AC_1-Vlanif100] dhcp select interface
[AC_1-Vlanif100] dhcp server excluded-ip-address 10.23.100.2
[AC_1-Vlanif100] quit
[AC_1] interface vlanif 101
[AC_1-Vlanif101] dhcp select interface
[AC_1-Vlanif101] dhcp server excluded-ip-address 10.23.101.2
```

```
[AC_1-Vlanif101] quit
在AC_1 上配置AP 上线
# 创建AP 组，用于将相同配置的AP 都加入同一AP 组中。
[AC_1] wlan
[AC_1-wlan-view] ap-group name ap1
[AC_1-wlan-ap-group-ap1] quit
# 创建域管理模板，在域管理模板下配置AC 的国家码并在AP 组下引用域管理模板。
[AC_1-wlan-view] regulatory-domain-profile name default
[AC_1-wlan-regulate-domain-default] country-code cn
[AC_1-wlan-regulate-domain-default] quit
[AC_1-wlan-view] ap-group name ap1
[AC_1-wlan-ap-group-ap1] regulatory-domain-profile default
Warning: Modifying the country code will clear channel, power and antenna gain configurations
of the radio and reset the AP. Continue?[Y/N]:y
[AC_1-wlan-ap-group-ap1] quit
[AC_1-wlan-view] quit
# 配置AC 的源接口。
[AC_1] capwap source interface vlanif 100
# 在AC 上离线导入AP，并将AP 加入AP 组“ap1”中
[AC_1] wlan
[AC_1-wlan-view] ap auth-mode mac-auth
[AC_1-wlan-view] ap-id 0 ap-mac 00e0-fc24-38f0
[AC_1-wlan-ap-0] ap-name area_1
[AC_1-wlan-ap-0] ap-group ap1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel,
power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC_1-wlan-ap-0] quit
# 将AP 上电后，当执行命令display ap all 查看到AP 的“State”字段为“nor”时，表示AP
正常上线。
```

```
<AC_1>display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor   : normal           [1]
-----
ID     MAC                Name    Group IP      Type           State STA Uptime
-----
0      00e0-fc24-38f0 area_1 ap1    10.23.100.243 AP4050DN-E     nor    0    55
-----
Total: 1
<AC_1>
```

```
# 创建名为“wlan-net”的安全模板，并配置安全策略。
[AC_1-wlan-view] security-profile name wlan-net
[AC_1-wlan-sec-prof-wlan-net] security wpa-wpa2 psk pass-phrase a1234567 aes
[AC_1-wlan-sec-prof-wlan-net] quit
# 创建名为“wlan-net”的SSID 模板，并配置SSID 名称为“wlan-net”。
[AC_1-wlan-view] ssid-profile name wlan-net
[AC_1-wlan-ssid-prof-wlan-net] ssid wlan-net
```

```

[AC_1-wlan-ssid-prof-wlan-net] quit
# 创建名为“wlan-net”的VAP 模板，配置业务数据转发模式、业务VLAN，并且引用安全模板和SSID 模板。
[AC_1-wlan-view] vap-profile name wlan-net
[AC_1-wlan-vap-prof-wlan-net] forward-mode tunnel
[AC_1-wlan-vap-prof-wlan-net] service-vlan vlan-id 101
[AC_1-wlan-vap-prof-wlan-net] security-profile wlan-net
[AC_1-wlan-vap-prof-wlan-net] ssid-profile wlan-net
[AC_1-wlan-vap-prof-wlan-net] quit
# 配置AP 组引用VAP 模板，AP 上射频0 和射频1 都使用VAP 模板“wlan-net”的配置。
WLAN 后面数字表示WLAN ID，表示其在VAP 内部编号。
[AC_1-wlan-view] ap-group name ap1
[AC_1-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 0
[AC_1-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 1
[AC_1-wlan-ap-group-ap1] quit
在AC_2 上配置AP 上线
# 创建AP 组，用于将相同配置的AP 都加入同一AP 组中。
[AC_2] wlan
[AC_2-wlan-view] ap-group name ap1
[AC_2-wlan-ap-group-ap1] quit
# 创建域管理模板，在域管理模板下配置AC 的国家码并在AP 组下引用域管理模板。
[AC_2-wlan-view] regulatory-domain-profile name default
[AC_2-wlan-regulate-domain-default] country-code cn
[AC_2-wlan-regulate-domain-default] quit
[AC_2-wlan-view] ap-group name ap1
[AC_2-wlan-ap-group-ap1] regulatory-domain-profile default
Warning: Modifying the country code will clear channel, power and antenna gain configurations
of the radio and reset the AP. Continue?[Y/N]:y
[AC_2-wlan-ap-group-ap1] quit
[AC_2-wlan-view] quit
# 配置AC 的源接口。
[AC_2] capwap source interface vlanif 100
# 在AC 上离线导入AP，并将AP 加入AP 组“ap1”中
[AC_2] wlan
[AC_2-wlan-view] ap auth-mode mac-auth
[AC_2-wlan-view] ap-id 0 ap-mac 00e0-fc48-2550
[AC_2-wlan-ap-0] ap-name area_2
[AC_2-wlan-ap-0] ap-group ap1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel,
power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC_2-wlan-ap-0] quit
# 将AP 上电后，当执行命令display ap all 查看到AP 的“State”字段为“nor”时，表示AP
正常上线。

```

```

<AC_2>display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor : normal [1]
-----
ID      MAC      Name      Group  IP      Type      State STA Uptime
-----
0       00e0-fc48-2550 area_2 ap1   10.23.100.55 AP4050DN-E nor    0    1M:37S
-----
Total: 1
<AC_2>|

```

在AC_2 上配置WLAN 业务参数

创建名为“wlan-net”的安全模板，并配置安全策略。

```
[AC_2-wlan-view] security-profile name wlan-net
```

```
[AC_2-wlan-sec-prof-wlan-net] security wpa-wpa2 psk pass-phrase a1234567 aes
```

```
[AC_2-wlan-sec-prof-wlan-net] quit
```

创建名为“wlan-net”的SSID 模板，并配置SSID 名称为“wlan-net”。

```
[AC_2-wlan-view] ssid-profile name wlan-net
```

```
[AC_2-wlan-ssid-prof-wlan-net] ssid wlan-net
```

```
[AC_2-wlan-ssid-prof-wlan-net] quit
```

创建名为“wlan-net”的VAP 模板，配置业务数据转发模式、业务VLAN，并且引用安全模板和SSID 模板。

```
[AC_2-wlan-view] vap-profile name wlan-net
```

```
[AC_2-wlan-vap-prof-wlan-net] forward-mode tunnel
```

```
[AC_2-wlan-vap-prof-wlan-net] service-vlan vlan-id 101
```

```
[AC_2-wlan-vap-prof-wlan-net] security-profile wlan-net
```

```
[AC_2-wlan-vap-prof-wlan-net] ssid-profile wlan-net
```

```
[AC_2-wlan-vap-prof-wlan-net] quit
```

配置AP 组引用VAP 模板，AP 上射频0 和射频1 都使用VAP 模板“wlan-net”的配置。

WLAN 后面数字表示WLAN ID，表示其在VAP 内部编号。

```
[AC_2-wlan-view] ap-group name ap1
```

```
[AC_2-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 0
```

```
[AC_2-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 1
```

```
[AC_2-wlan-ap-group-ap1] quit
```

4.1.4 配置 AC_1 的 WLAN 漫游功能

创建漫游组，并配置AC_1 和AC_2 为漫游组成员。

```
[AC_1-wlan-view] mobility-group name mobility
```

```
[AC_1-mc-mg-mobility] member ip-address 10.23.100.1
```

```
[AC_1-mc-mg-mobility] member ip-address 10.23.100.2
```

```
[AC_1-mc-mg-mobility] quit
```

4.1.5 配置 AC_2 的 WLAN 漫游功能

创建漫游组，并配置AC_1 和AC_2 为漫游组成员。

```
[AC_2-wlan-view] mobility-group name mobility
```

```
[AC_2-mc-mg-mobility] member ip-address 10.23.100.1
```

```
[AC_2-mc-mg-mobility] member ip-address 10.23.100.2
```

```
[AC_2-mc-mg-mobility] quit
```

4.2 测试网络

查看漫游组配置

```
[AC_2-wlan-view]display mobility-group name mobility
-----
State          IP address          Description
-----
normal         10.23.100.1         -
normal         10.23.100.2         -
-----
Total: 2
[AC_2-wlan-view]
```

查看 sta 漫游轨迹

在 AC_2 上执行命令 display station roam-track sta-mac e019-1dc7-1e08，可以查看该 STA 的漫游轨迹。

```
[AC_2-wlan-view] display station roam-track sta-mac e019-1dc7-1e08
```

Access SSID:wlan-net

Rx/Tx: link receive rate/link transmit rate(Mbps)

z:Zero Roam c:PMK Cache Roam r:802.11r Roam

L2/L3 BSSID	AC IP TIME	AP name In/Out RSSI	Radio ID Out Rx/Tx
--	10.23.100.1	area_1	1
60de-4476-e360	2019/03/09 16:11:51	-57/-57	22/3
L2	10.23.100.2	area_2	1
dcd2-fc04-b500	2019/03/09 16:13:53	-58/-	-/-

Number: 1

Ensp 上可能看不到两个 ap name